DATABASE DESIGN & IMPLEMENTATION DOCUMENT

NAILVERSE

Not just NAILS.. IT's the WHOLE MOOD
Online Nail Art & Nail Beauty Online Platform

WEEK – 3

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3. Supplier and Inventory Management System

Scenario

Nailverse sources nail-care products — gel polishes, acrylic powders, extension tips, and cuticle oils — from multiple suppliers so technicians always have what they need for a booked service. The system should maintain supplier details, track which products each supplier provides, and monitor inventory levels to ensure product availability before an appointment is confirmed.

1. **Create Supplier, Product and Inventory Tables**

```
SQL> CREATE TABLE Supplier(Supplier_ID VARCHAR2(7), Supplier_Nam
e VARCHAR2(25),Email VARCHAR2(30), Phone NUMBER(10), Address VAR
CHAR2(45));

Table created.

SQL> CREATE TABLE Product(Product_ID VARCHAR2(7), Product_Name V
ARCHAR2(30), Product_Type VARCHAR2(20));

Table created.
SQL> CREATE TABLE Inventory (
  2      Inventory_ID VARCHAR2(7),
  3      Product_ID VARCHAR2(7),
  4      Supplier_ID VARCHAR2(7),
  5      Stock_Quantity NUMBER,
  6      Stock_Status VARCHAR2(11),
  7      Last_Updated DATE
  8  );

Table created.
```

2. Establish Relationships

Supplier and Product

One supplier can supply multiple products.
One product can be supplied by one or multiple suppliers.
Relationship:
Supplier → Product is one to many

Product and Inventory
Each product has inventory information.
Inventory tracks available stock.
Relationship:
Product → Inventory is one to one
Primary Key added for all three tables:

```
SQL> ALTER TABLE Supplier ADD CONSTRAINT PK_Supplier PRIMARY KEY
(Supplier_ID);

Table altered.

SQL> ALTER TABLE Product ADD CONSTRAINT PK_Product PRIMARY KEY(P
roduct_ID);

Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inv
entory_ID);

Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT FK_Prod FOREIGN KEY(Pr
oduct_ID) REFERENCES Product(Product_ID);

Table altered.
SQL> ALTER TABLE Inventory
  2  ADD CONSTRAINT FK_Supplier
  3  FOREIGN KEY (Supplier_ID)
  4  REFERENCES Supplier(Supplier_ID);

Table altered.
```

Constraints to make Unique Email ID:

```
SQL> ALTER TABLE Supplier ADD CONSTRAINT Un_mail UNIQUE(Email);

Table altered.
```

To make Email NOT NULL

```
SQL> ALTER TABLE Supplier MODIFY Email NOT NULL;

Table altered.
```

Add Constraint to Check the Stock Quantity is Positive

```
SQL> ALTER TABLE Inventory ADD CONSTRAINT CK_Quan CHECK(Stock_Qu
antity >=0);

Table altered.
```

3. Insert Values

Insert Values to Supplier Table

```
SQL> INSERT INTO Supplier VALUES('&SupID','&SupplierName','&emai
l',&phone,'&Address');
Enter value for supid: S00001
Enter value for suppliername: GLAMNAIL SUPPLIES
Enter value for email: glamnailsupplies@gmail.com
Enter value for phone: 8765432109
Enter value for address: 1, Anna Nagar, Chennai
old   1: INSERT INTO Supplier VALUES('&SupID','&SupplierName','&
email',&phone,'&Address')
new   1: INSERT INTO Supplier VALUES('S00001','GLAMNAIL SUPPLIES
','glamnailsupplies@gmail.com',8765432109,'1, Anna Nagar, Chenna
i')

1 row created.

SQL> select * from Supplier;

SUPPLIE SUPPLIER_NAME             EMAIL                          PHONE
------- ------------------------- ------------------------------ ----------
ADDRESS
----------------------------------------
S00001  GLAMNAIL SUPPLIES         glamnailsupplies@gmail.com    8765432109
1, Anna Nagar, Chennai

S00002  POLISH POINT Co.          polishpointco@gmail.com       9876543012
90, George Town, Chennai

S00003  ARTNAIL TRADERS           artnailtraders@gmail.com      7890654312
62, Thottapalayam, Vellore


SUPPLIE SUPPLIER_NAME             EMAIL                          PHONE
------- ------------------------- ------------------------------ ----------
ADDRESS
----------------------------------------
S00004  SHINECRAFT DISTRIBUTOR    shinedistributor@gmail.com     784003009
23, Thillai Nagar, Trichy

S00005  BADDIES NEED NAILS        baddiesneednails@gmail.com    6789904321
143, Lakshmi Nagar, Chennai
```

Insert Values to Product Table

```
SQL> INSERT INTO Product VALUES('&ProdID','&ProductName','&ProdType');
Enter value for prodid: P100001
Enter value for productname: GEL POLISH - RUBY RED
Enter value for prodtype: POLISH
old   1: INSERT INTO Product VALUES('&ProdID','&ProductName','&ProdType')
new   1: INSERT INTO Product VALUES('P100001','GEL POLISH - RUBY RED','POLI
SH')

1 row created.
```

```
SQL> select * from Product;

PRODUCT PRODUCT_NAME                  PRODUCT_TYPE
------- ----------------------------- --------------------
P100001 GEL POLISH - RUBY RED         POLISH
P100002 ACRYLIC POWDER - CLEAR        ACRYLIC
P100005 GEL POLISH - LIGHT LIME       POLISH
P100007 NAIL EXTENSION TIP - ALMOND   EXTENSION
P200008 CUTICLE OIL - LAVENDER        CARE
P200017 CUTICLE OIL - PEACH           CARE

6 rows selected.
```

Insert Values to Inventory Table

```
SQL> INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sup', &Stock,'&Stock
_status',TO_DATE('&Last_upd','YYYY-MM-DD'));
Enter value for invid: INV0001
Enter value for prodid: P100001
Enter value for sup: S00001
Enter value for stock: 5
Enter value for stock_status: AVAILABLE
Enter value for last_upd: 12-SEP-26
old   1: INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sup', &Stock,'&S
tock_status',TO_DATE('&Last_upd','YYYY-MM-DD'))
new   1: INSERT INTO Inventory VALUES('INV0001','P100001','S00001', 5,'AVAI
LABLE ',TO_DATE('12-SEP-26','YYYY-MM-DD'))

1 row created.
```

```
SQL> SELECT * FROM Inventory;

INVENTO PRODUCT SUPPLIE STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ---------- ---------
INV0001 P100001 S00001               5 AVAILABLE  26-SEP-12
INV0002 P100001 S00002               4 AVAILABLE  26-OCT-13
INV0003 P100002 S00002               2 UNAVAILABLE 26-FEB-14
INV0004 P200017 S00004               4 AVAILABLE  26-AUG-09
```

Report 1: Supplier-wise Product Report
Display:

- Supplier name
- Number of products supplied
- Available stock

```
SQL> SELECT s.Supplier_Name,
  2         COUNT(i.Product_ID) AS Number_of_Products,
  3         SUM(i.Stock_Quantity) AS Available_Stock
  4  FROM Supplier s
  5  JOIN Inventory i
  6  ON s.Supplier_ID = i.Supplier_ID
  7  WHERE UPPER(i.Stock_Status) = 'AVAILABLE'
  8  GROUP BY s.Supplier_Name;

SUPPLIER_NAME             NUMBER_OF_PRODUCTS AVAILABLE_STOCK
------------------------- ------------------ ---------------
SHINECRAFT DISTRIBUTOR                     1               4
POLISH POINT Co.                           1               4
```

Use Joins to get details from Supplier and Inventory Table.

Report 2: Stock Availability Report
Display:

- Product name
- Stock quantity
- Stock status

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status FRO
M Product p JOIN Inventory i ON p.Product_ID = i.Product_ID;

PRODUCT_NAME                   STOCK_QUANTITY STOCK_STATU
------------------------------ -------------- -----------
GEL POLISH - RUBY RED                       5 AVAILABLE
GEL POLISH - RUBY RED                       4 AVAILABLE
ACRYLIC POWDER - CLEAR                      2 UNAVAILABLE
CUTICLE OIL - PEACH                         4 AVAILABLE
```

Report 3: Inventory Analysis
Find:

- Total products available

```
SQL>  SELECT COUNT(*) AS Total_Available_Products FROM Inventory
 WHERE Stock_Status = 'Available';

TOTAL_AVAILABLE_PRODUCTS
------------------------
                       0
```

- Products out of stock

```
SQL> SELECT COUNT(*) AS OutOfStock_Products FROM Inventory WHERE
 Stock_Status = 'Out of Stock';

OUTOFSTOCK_PRODUCTS
-------------------
                  0
```

- Highest stocked products

```
SQL> SELECT p.Product_Name, i.Stock_Quantity FROM Product p JOIN
 Inventory i ON p.Product_ID = i.Product_ID WHERE i.Stock_Quanti
ty = (SELECT MAX(Stock_Quantity) FROM Inventory);

PRODUCT_NAME                   STOCK_QUANTITY
------------------------------ --------------
GEL POLISH - RUBY RED                       5
```

- Average inventory quantity

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory FROM Invent
ory;

AVERAGE_INVENTORY
-----------------
             3.75
```